

Nikhil Vidhani

Lead Data Scientist – Simulation, Optimisation & Agentic AI

Executive Summary

- Architected and now lead the technical evolution of Nemo, Target's end-to-end simulation platform for order allocation and available-to-promise, used as the primary testbed for network policy design, with decisions potentially affecting up to \$8B in digital sales.
- Architected and deployed Nemo AI, a production multi-agent research and diagnosis system, while introducing reusable Model Context Protocol (MCP)-based capabilities and AI-engineering practices for evaluation, traceability, observability, and responsible adoption.
- Conceived and led the redesign of Target's legacy consolidation algorithm, translating customer-speed and shipping-cost trade-offs into a principled optimisation framework now in A/B-test preparation, with projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental sales.
- Deep technical expertise in Python and R, distributed processing (PySpark, Kafka, Hive/Hadoop, and SQL), simulation, operations research, algorithm design, machine learning, and production software engineering, supported by a Ph.D. in Finance from IIM Bangalore and a master's degree in engineering from IISc Bangalore.
- Recognised as a go-to simulation and fulfilment expert, influencing product, engineering, operations, and data-science decisions while mentoring a five-person team through architecture, experimentation, code and design reviews, and technical communication; previously led a five-member analytics team at WNS.

Industry and Research Experience

Nov 2022 – **Lead Data Scientist**, *Target Corporation, Bangalore*, Team Size: 5.

- Present
- Own the technical strategy, architecture, and expansion of end-to-end simulation capabilities for order allocation and available-to-promise, setting direction across algorithms, data systems, validation, interfaces, and production operations.
 - Lead complex simulation and optimisation work on consolidation, promise speed, capacity, and fulfilment policy, converting ambiguous business questions into objective functions, scenarios, sensitivity analyses, and controlled experiments.
 - Establish platform-quality mechanisms including large-scale backtesting, automated accuracy measurement, common data and date engines, checkpoint recovery, self-service validation, and release controls to improve fidelity, reliability, and developer productivity.
 - Serve as *AI Champion* for the Digital Fulfilment data-science organisation—architecting production agentic-AI systems and reusable MCP/skill capabilities, and setting practices for specification-driven development, testing, prompt versioning, observability, and evaluation.
 - Partner with product, engineering, supply-chain, and fulfilment leaders as a go-to technical expert; shape roadmaps and launch decisions while mentoring data scientists on architecture, experimentation, debugging, and stakeholder communication.

- Feb 2022 – **Sr. Group Manager (Data Science)**, *WNS Global Services, Bangalore*, Team Size: 5.
- Nov 2022
- Led a five-member data-science team delivering analytics and machine-learning solutions for finance and accounting use cases (payments, collections, reporting, prediction, and auditing) for global clients, driving multi-million-dollar operational savings.
 - Owned end-to-end technical delivery, from problem framing with business stakeholders through data engineering, modelling, API development, and automated deployment, ensuring reliable and maintainable solutions.
 - Established shared GitHub repositories, code-quality standards, and CI/CD workflows; raised team capability through code reviews, technical sessions, and structured knowledge-sharing.
- Jun 2016 – **Research Scholar**, *Indian Institute of Management, Bangalore*, IC.
- Jan 2022
- Researched asset-pricing anomalies and examined the impact of disagreement on trading volume, analyst forecasts, and 10-K document characteristics.
 - Conducted comprehensive data collection, cleaning, modelling, and statistical analysis.
 - Published multiple papers and presented findings at several conferences. *Complete list available at: publications_CV.pdf*
- Jun 2018 – **Primary Instructor**, *Programming and Data Analysis*, Batch Size: 5–40.
- Jan 2022
- Designed and taught multiple short- and long-format courses in R programming, data analysis, and LaTeX.
 - Instructor rating: 4.7/5 (<https://github.com/nik141088/applied-R>).
- Jul 2012 – **Software Engineer**, *Cisco Systems, Bangalore*, IC.
- May 2016
- Designed, implemented, tested, reviewed, and documented software for 4G-LTE and Wi-Fi systems.
 - Contributed to high-availability, redundancy, and network-security architecture.

Major Projects

- 2026 – **Nemo AI & Generative AI Enablement**, *Target*.
- Present
- Conceived, architected, built, and deployed **Nemo AI**, an internally hosted multi-agent orchestrator/executor/synthesiser that performs deep research on simulation runs, answers conceptual questions, and analyses inputs and outputs to accelerate root-cause diagnosis.
 - Engineered production-grade capabilities across local, development, and production environments, including authentication, per-request cost and latency tracking, persisted history, and full traces for debugging, observability, governance, and evaluation.
 - Designed a specification-driven “LLM Wiki” approach—an alternative to retrieval-augmented generation (RAG) for medium-sized knowledge bases—with hierarchical curation and immutable citations back to source documents to improve traceability and reduce hallucinations.
 - Built a Model Context Protocol (MCP) service over core diagnostic systems and reusable skills and agents for simulation root-cause analysis, code review, and PR generation, reducing repetitive manual analysis and enabling reuse beyond a single application.
 - Led weekly AI-engineering forums and share-outs, translating emerging practices into standards for specifications, testing, prompt versioning, golden-set evaluation, observability, and human ownership of AI-generated code.

2025 – Present	Consolidation Algorithm Revamp, Target. ○ Conceived and redesigned Target’s consolidation algorithm to jointly optimise customer promise speed and shipping cost across the last-mile network, replacing legacy heuristics with a principled modelling framework. ○ Formulated the cost-versus-speed objective and used Nemo’s integrated promise and order-allocation simulations for scenario and sensitivity analysis across realistic order, inventory, capacity, and No-Rush conditions. ○ Led rigorous validation across data-science and engineering implementations, increasing functional agreement from 20% to 95% and identifying defects in both; progressed to mirror testing and designed a controlled A/B test across 10 ZIP codes, with speed, cost, and No-Rush offer rate as core metrics. ○ Quantified projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental digital sales through improved conversion and margin. ○ Implemented, profiled, tested, and tuned the algorithm for a latency-critical promise service, balancing model quality with strict performance and reliability constraints. ○ Served as the technical bridge across product, engineering, and fulfilment through decision documents, design reviews, curated test cases, presentations, and knowledge transfer, enabling phased rollout planning.
2023 – Present	Nemo: Unified Simulation Platform, Target. ○ Architected and led a five-member team in building and operating Nemo, a unified simulation platform for last-mile promise and fulfilment, supporting more than 100 stakeholders across planning, digital fulfilment, and operations. ○ Developed an available-to-promise (ATP) simulator from scratch—spanning algorithm design, data modelling, interfaces, and deployment automation—and integrated it with the guest order allocation (GOA) simulator to evaluate end-to-end speed, node selection, capacity, and shipping-cost trade-offs. ○ Established large-scale backtesting and automated accuracy pipelines, improving ATP accuracy from about 40% to 85%; later diagnosed a systemic time-zone defect that raised before/after-cutoff agreement from 39% to 88%. ○ Modernised the platform with Kafka-based ingestion, scalable PySpark/Hive processing, common data and date engines, input validation, checkpoint recovery, and automated reporting; improved runtime by 40% through exhaustive profiling and optimisation. ○ Delivered simulation studies informing decisions affecting roughly 3% of Target’s digital orders (\$600M+), including an evaluation that identified up to \$5M in potential shipping-cost savings for an estimated 0.03-day increase in delivery time. ○ Owned the technical roadmap across five major and 12 minor releases in 2025–2026, spanning accuracy, replay, self-service, interfaces, and future cloud/agent architecture; presented “Scaling Nemo: Experiences and Challenges” at an organisation-wide Simulations Conference.
2021 – 2024	Past Projects, Target, WNS, IIMB, and Personal. <i>A list of selected past projects: selected_past_projects_CV.pdf</i>

Tools and Skills

Expert	Programming & Data Engineering , Python, R, PySpark, SQL, Kafka, Hive/Hadoop, distributed data pipelines, data modelling, Shiny/Streamlit, C/C++.
Advanced	Analytics & Modelling , Simulation, operations research, local-search and heuristic optimisation, experiment design, A/B testing, regression and classification, NLP, feature engineering, TensorFlow/Keras.
Advanced	Generative AI & Agentic AI , Multi-agent orchestration, tool calling, Model Context Protocol (MCP), LLM application architecture, prompt/skill/agent engineering, evaluation, traceability, governance, observability, specification-driven development.
Expert	Design Principles , System and algorithm design, data structures, time and space complexity, performance profiling, vectorisation, functional programming, test-driven and modular development, secure coding.

Advanced	Engineering & Platforms , APIs and services (Plumber/FastAPI), distributed processing, authentication, observability, debugging, Linux, code review, Git/GitHub, relational databases, AWS, cloud-migration planning, CI/CD.
Advanced	Visualisation & Storytelling , Decision documents, technical presentations, ggplot2, Plotly, R Markdown, dashboarding, LaTeX, Confluence/Jira.
Expert	Domain Knowledge , Last-mile operations research (available-to-promise and order allocation), digital fulfilment, promise-speed and shipping-cost optimisation, finance, accounting, and statistics.
Advanced	Product & Strategy , Technical strategy, architecture and roadmap design, ambiguous problem framing, scenario design, impact sizing, product thinking, and cross-functional decision support.
Advanced	Technical Leadership , Technical mentoring and coaching, architecture and code reviews, engineering standards, knowledge transfer, project leadership, capability development, stakeholder influence, and candidate interviewing.

Education

2016 – 2021	Doctor of Philosophy , <i>Finance and Accounting</i> , Indian Institute of Management, Bangalore. CGPA: 3.64 / 4.00
2010 – 2012	Master of Engineering , <i>Electrical Communication Engineering</i> , Indian Institute of Science, Bangalore. CGPA: 6.4 / 8.0; Project Grade: A
2006 – 2010	Bachelor of Technology , <i>Electronics and Communication Engineering</i> , Bundelkhand Institute of Engineering and Technology, Jhansi (U.P.). Grade: 73.6%
2004 – 2006	Schooling (12th Standard) , <i>Science</i> , C.B.S.E.. Grade: 88.2%

Awards and Honours

2025	Drive , Target Corporation.
2022	Tech Genius, Transforming the Organization, Victory Fleet , WNS Global Services.
2020	Mirae Asset Scholarship (PhD Year 5) , Indian Institute of Management.
2017–2018	Director's Merit List (PhD Years 1 and 2) , Indian Institute of Management.
2016	96.5th percentile , Common Aptitude Test (CAT).
2010	All India Rank 20 of 105,000 , Graduate Aptitude Test in Engineering (GATE).
2006–2007	Merit Scholarships , Intermediate Examination and Engineering.